Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate cross correlation and autocorrelation on discrete time signals.**	
Experiment:-3	**Date:02-08-2026**	**Enrollment No:92400133063**

➢ Objective:-

- To understand the mathematical definitions of autocorrelation and cross-correlation.
- To simulate these operations on deterministic and random discrete signals using Python.
- To interpret the results for signal similarity, periodicity, and time-delay estimation.

➢ Theory

- What is Correlation?

Correlation is a measure of similarity between two signals as a function of a time-lag applied to one of them. It is widely used in radar, sonar, communications (synchronization), and system identification.

- Autocorrelation

The autocorrelation function of a discrete-time signal $\boldsymbol{x[n]}$ measures the similarity between the signal and a time-shifted version of itself. For a finite-length signal of length $\boldsymbol{N}$, it is defined as:

$$\boldsymbol{r_{xx}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n-l]}$$

➢ where $\boldsymbol{l}$ is the lag (integer shift).
Interpretation:

- At $\boldsymbol{l = 0}$, $\boldsymbol{r_{xx}[0] = \sum x[n]^2}$, which is the total energy of the signal.
- Autocorrelation is even symmetric: $\boldsymbol{r_{xx}[l] = r_{xx}[-l]}$.
- The maximum always occurs at zero lag: $\boldsymbol{| r_{xx}[l] | \leq r_{xx}[0]}$.

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➢ Cross-Correlation:-

The cross-correlation between two different signals $\boldsymbol{x[n]}$ and $\boldsymbol{y[n]}$ measures how well $\boldsymbol{y[n]}$ matches $\boldsymbol{x[n]}$ when shifted:

$$\boldsymbol{r_{xy}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n-l]}$$

➢ Interpretation:

- If $\boldsymbol{y[n] = x[n-d]}$ (i.e., a delayed version of $\boldsymbol{x}$), then $\boldsymbol{r_{xy}[l]}$ will have its peak at $\boldsymbol{l = d}$.
- Cross-correlation is not generally symmetric.

➢ Normalized Correlation (Correlation Coefficient)

Often we normalize to get values between -1 and 1 for easier interpretation:

$$\boldsymbol{\rho_{xy}[l] = \frac{r_{xy}[l]}{\sqrt{r_{xx}[0] \cdot r_{yy}[0]}}}$$

where $\boldsymbol{\rho = 1}$ means identical signals (for that lag), $\boldsymbol{\rho = -1}$ means opposite polarity, and $\boldsymbol{\rho = 0}$ means completely unrelated.

➢ Applications in DSP

- Time-delay estimation: Finding the lag where cross-correlation peaks gives the time shift between two received signals (e.g., in target ranging).
- Periodicity detection: Autocorrelation of a periodic signal has peaks at multiples of the period.
- Random signal analysis: White noise has an autocorrelation that is zero everywhere except at lag 0 (ideal for decorrelation).

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1. Normalized cross-correlation:-

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

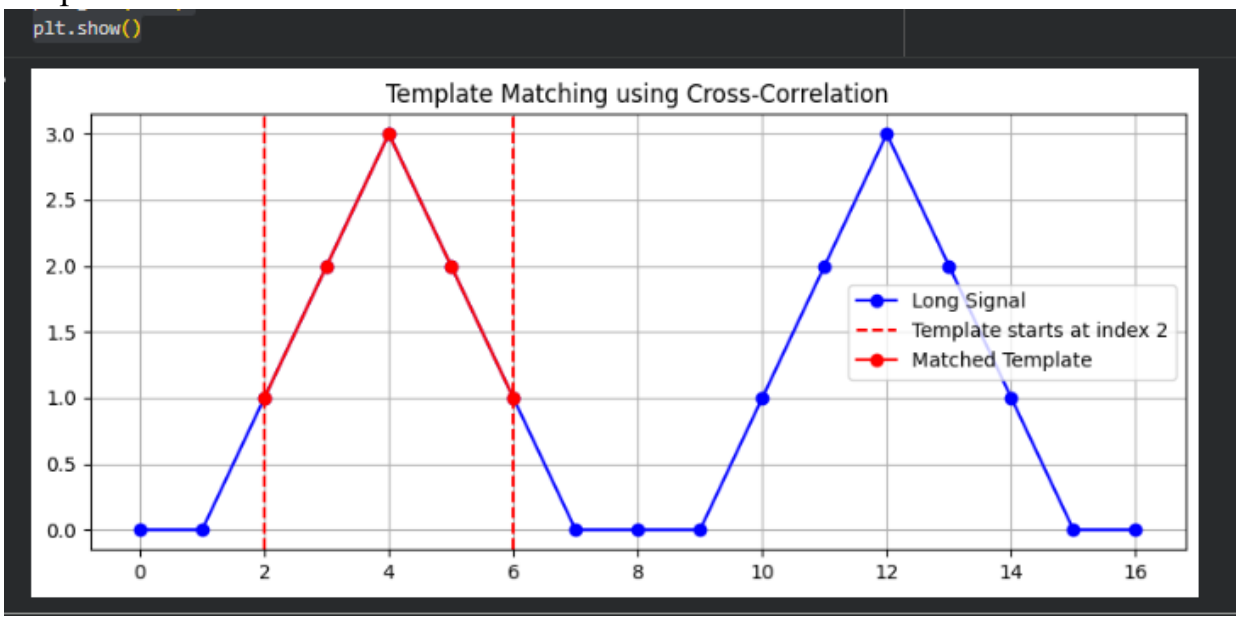
# Long signal
long_signal = np.array([0, 0, 1, 2, 3, 2, 1, 0, 0, 0, 1, 2, 3, 2, 1, 0, 0])
# Template to find
template = np.array([1, 2, 3, 2, 1])

# Compute cross-correlation
corr = np.correlate(long_signal, template, mode='valid')
peak_idx = np.argmax(corr)

plt.figure(figsize=(10, 4))
plt.plot(long_signal, 'bo-', label='Long Signal')
plt.axvline(peak_idx, color='red', linestyle='--', label=f'Template starts at index
{peak_idx}')
plt.axvline(peak_idx + len(template) - 1, color='red', linestyle='--')
plt.plot(np.arange(peak_idx, peak_idx + len(template)), template, 'ro-',
label='Matched Template')
plt.legend()
plt.title('Template Matching using Cross-Correlation')
plt.grid(True)
plt.show()
```

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Output:-



2. **Cross-Correlation to Detect Time Delay**

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

# Generate a random signal
np.random.seed(42)
x = np.random.randn(50)
# Create a delayed version (delay = 10 samples)
delay = 10
y = np.zeros_like(x)
y[delay:] = x[:-delay]  # y[n] = x[n-10]
# Add a tiny bit of noise to make it realistic
y = y + 0.05 * np.random.randn(50)

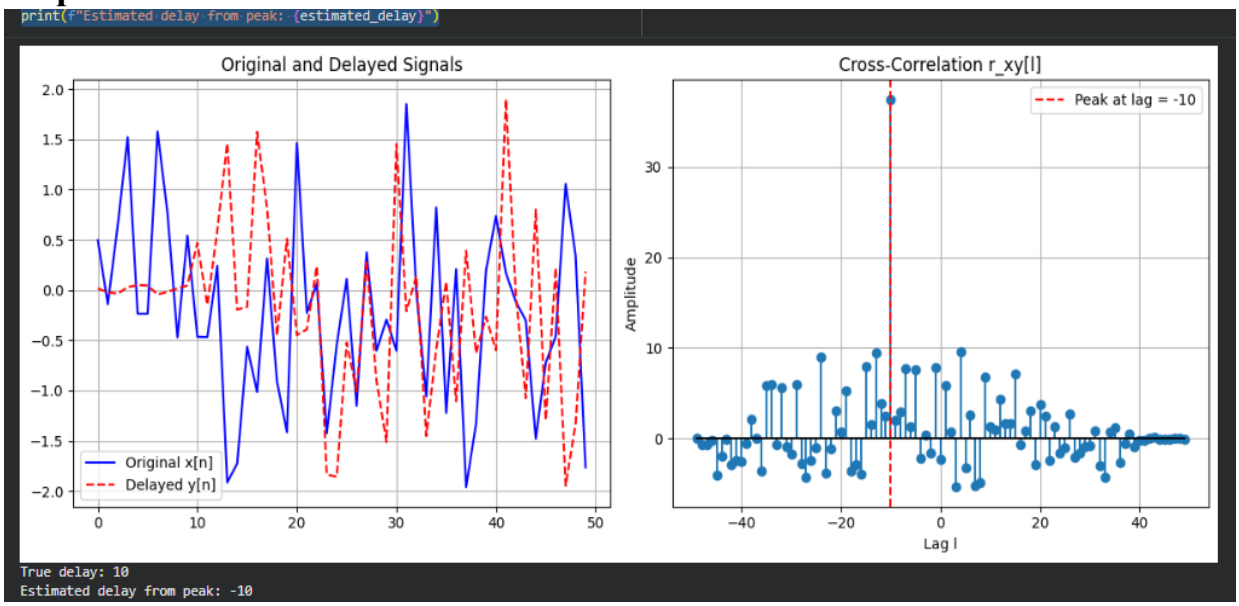
# Cross-correlation
cross_corr = np.correlate(x, y, mode='full')
```

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```
# Lags for cross-correlation: length = len(x) + len(y) - 1 = 99
Nx = len(x)
Ny = len(y)
lags = np.arange(-Ny + 1, Nx)
# Find the peak lag
peak_idx = np.argmax(cross_corr)
estimated_delay = lags[peak_idx]
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(x, 'b-', label='Original x[n]')
plt.plot(y, 'r--', label='Delayed y[n]')
plt.legend()
plt.title('Original and Delayed Signals')
plt.grid(True)
plt.subplot(1, 2, 2)
plt.stem(lags, cross_corr, basefmt="k-")
plt.axvline(estimated_delay, color='red', linestyle='--', label=f'Peak at lag =
{estimated_delay}')
plt.title('Cross-Correlation r_xy[l]')
plt.xlabel('Lag l')
plt.ylabel('Amplitude')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
print(f"True delay: {delay}")
print(f"Estimated delay from peak: {estimated_delay}")
```

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output:-



3. Autocorrelation of a Deterministic Signal

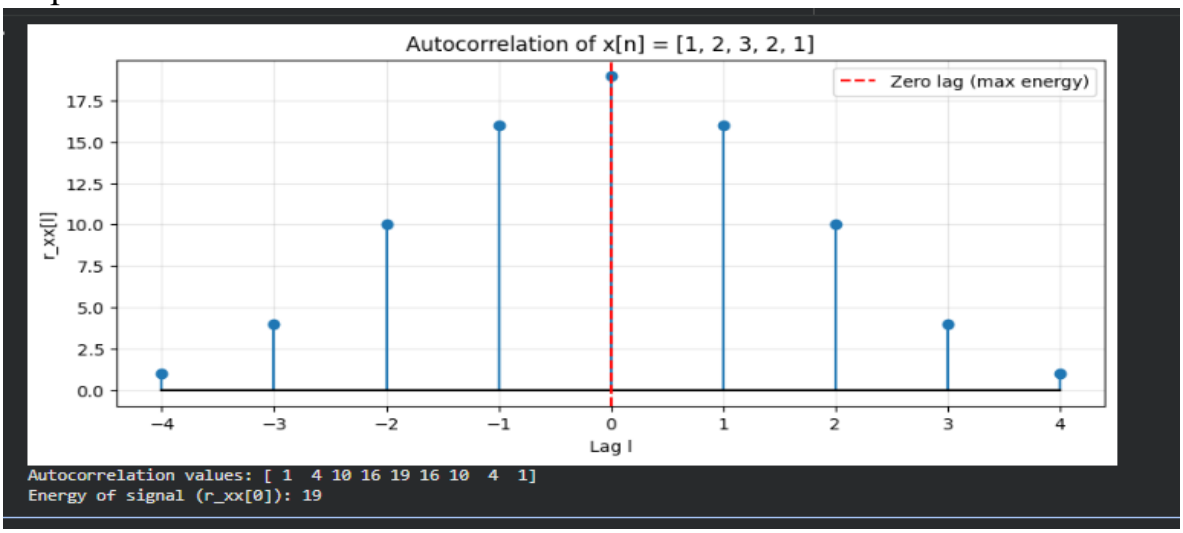
Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Define a simple deterministic signal
x = np.array([1, 2, 3, 2, 1])
# Compute autocorrelation using numpy.correlate (mode='full' preserves all lags)
auto_corr = np.correlate(x, x, mode='full')
# Create lag axis. For length N, lags go from -(N-1) to (N-1)
N = len(x)
lags = np.arange(-N + 1, N)
# Plot
plt.figure(figsize=(10, 4))
plt.stem(lags, auto_corr, basefmt="k-")
plt.title('Autocorrelation of x[n] = [1, 2, 3, 2, 1]')
plt.xlabel('Lag l')
```

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```
plt.ylabel('r_xx[l]')
plt.grid(True, alpha=0.3)
plt.axvline(0, color='red', linestyle='--', label='Zero lag (max energy)')
plt.legend()
plt.show()
print("Autocorrelation values:", auto_corr)
print("Energy of signal (r_xx[0]):", auto_corr[N-1])
```

output:-



4. **Autocorrelation – Periodic vs. Random Signals**

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(0, 200)
# Periodic signal: Sine wave
fs = 1
f = 0.05  # 5 cycles in 100 samples
periodic_signal = np.sin(2 * np.pi * f * n)
```

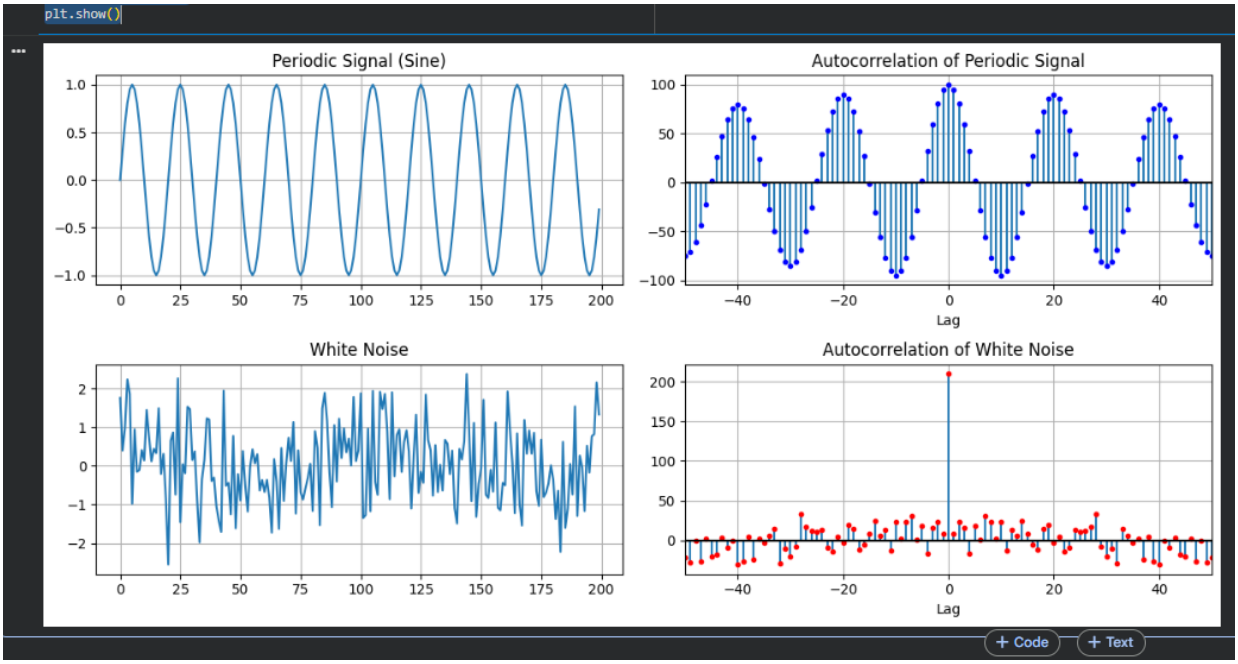
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```
# Random signal (White noise)
np.random.seed(0)
noise_signal = np.random.randn(200)
# Compute autocorrelations
auto_periodic = np.correlate(periodic_signal, periodic_signal, mode='full')
auto_noise = np.correlate(noise_signal, noise_signal, mode='full')
N = len(periodic_signal)
lags = np.arange(-N + 1, N)
plt.figure(figsize=(12, 6))
plt.subplot(2, 2, 1)
plt.plot(n, periodic_signal)
plt.title('Periodic Signal (Sine)')
plt.grid(True)
plt.subplot(2, 2, 2)
plt.stem(lags, auto_periodic, basefmt="k-", markerfmt='b.')
plt.title('Autocorrelation of Periodic Signal')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.subplot(2, 2, 3)
plt.plot(n, noise_signal)
plt.title('White Noise')
plt.grid(True)

plt.subplot(2, 2, 4)
plt.stem(lags, auto_noise, basefmt="k-", markerfmt='r.')
plt.title('Autocorrelation of White Noise')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.tight_layout()
plt.show()
```

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Conclusion:-

- **Autocorrelation** confirmed symmetry and peaked at zero lag, accurately representing signal energy.

- **Cross-correlation** effectively estimated time delays, locating the peak at the correct shift even in the presence of noise.

- **Periodic and random signals** were clearly distinguishable—periodic signals showed sustained oscillatory correlations, while white noise produced an impulse-like response with near-zero values elsewhere.

- **Template matching** via cross-correlation successfully identified the position of a known pattern within a longer signal.